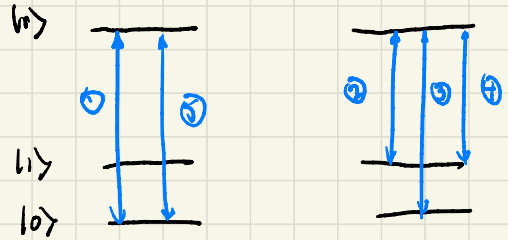
Question #1

Scheme:



1.1

Qubit basis $|0_c 0_t\rangle$

Pulse 1: $e^{i\pi/2} |r_c 0_t\rangle$

Pulse 2: $e^{i\pi/2} |r_c 0_t\rangle$ (no effect due to detuning)

Pulse 3: $e^{i\pi/2} |r_c 0_t\rangle$ (no effect due to blockade)

Pulse 4: $e^{i\pi/2} |r_c 0_t\rangle$ (no effect due to detuning)

Pulse 5: $e^{i\pi} |0_c 0_t\rangle = |0_c 0_t\rangle$

Qubit basis $|0_c 1_t\rangle$

Pulse 1: $e^{i\pi/2} |r_c 1_t\rangle$

Pulse 2: $e^{i\pi/2} |r_c 1_t\rangle$ (no effect due to blockade)

Pulse 3: $e^{i\pi/2} |r_c 1_t\rangle$ (no effect due to detuning)

Pulse 4: $e^{i\pi/2} |r_c 1_t\rangle$ (blockaded)

Pulse 5: $e^{i\pi} |0_c 1_t\rangle = |0_c 1_t\rangle$

Qubit basis $|1_c 0_t\rangle$

Pulse 1: $|1_c 0_t\rangle$ (detuned)

Pulse 2: $|1_c 0_t\rangle$ (detuned)

Pulse 3: $e^{-i\pi/2}|1_c r_t\rangle$ (resonant)

Pulse 4: $e^{i\pi}|1_c 1_t\rangle$ (resonant)

Pulse 5: $e^{i\pi}|1_c 1_t\rangle = |1_c 1_t\rangle$

Qubit basis $|1_c 1_t\rangle$

Pulse 1: $|1_c 1_t\rangle$ (detuned)

Pulse 2: $e^{i\frac{\pi}{2}}|1_c r_t\rangle$ (resonant)

Pulse 3: $e^{-i\pi}|1_c 0_t\rangle$

Pulse 4: $e^{i\pi}|1_c 0_t\rangle$ (detuned)

Pulse 5: $e^{i\pi}|1_c 0_t\rangle$ (detuned) $= |1_c 0_t\rangle$

1.2 From the above behavior we see that the overall matrix will be:

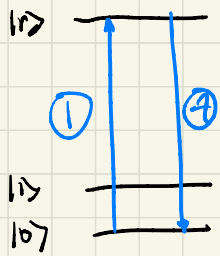
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \text{CNOT}.$$

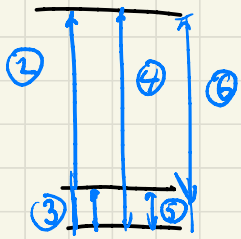
1.3 A key disadvantage is the use of multiple Rydberg excitation frequencies. Since there are optical transitions, this necessitates additional lasers, vs. just using the existing single qubit rotations between the two qubit states.

Any reasonable answer here could be accepted. Another disadvantage is the relative stability and fidelity of the single qubit rotations in comparison to the Rydberg transitions used here.

Advanced

a)





State $|0_c 0_t\rangle$ pulses 2 – 6 have no effect.

pulses 3 + 5 result in -1; pulses 1 + 7 result in -1

$\longrightarrow +|0_c 0_t\rangle$

State $|1_c 0_t\rangle$

pulses 1 + 7 no effect.

pulses 2, 4, 5 $e^{-i\pi/2}|1_c 1_t\rangle = -i|1_c 1_t\rangle$

State $|0_c 1_t\rangle$

$\longrightarrow |0_c 1_t\rangle$ (similar to $|0_c 1_t\rangle$ case)

State $|1_c 1_t\rangle$

$\longrightarrow$ pulse 2 no effect.
pulse 3 flip to $|0_t\rangle$; $e^{i\pi/2}$
pulse 4 to $|r_t\rangle$ $e^{i\pi}$
pulse 5, no effect.
pulse 6, $e^{i\pi/2}|1_c 0_t\rangle = -i|1_c 0_t\rangle$

Resulting matrix:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \end{bmatrix}$$

$\longrightarrow$ CNOT w/ additional single qubit phase.

Advanced

b) To realize a canonical CNOT, we need to rotate the state. We can use the P gate (called this the π-gate in class)

$$Z(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\delta} \end{pmatrix}$$

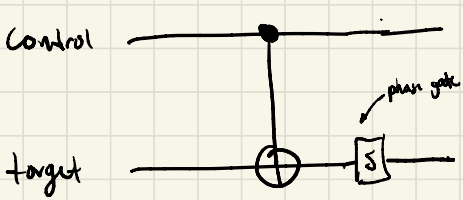
Set $\delta = +\pi/2$; resulting in

$$Z(\pi/2) = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = S \text{ (phase gate)}$$

We apply this to the target qubit.

so that $Z(\pi/2)|1_t\rangle = i|1_t\rangle$; $|0_t\rangle \rightarrow |0_t\rangle$

The overall circuit is



Resulting matrix is

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & 0 & 0 & -i \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & i \\ 0 & 0 & i & 0 \end{bmatrix}$$

as required.